L1.1 for the canonical `dirac4` rule: the physical lattice is one BCC orbit (r2)

Quantized Space-Ledger. Written 12 November 2026, America/New_York. This note records a lattice choice made by Edmond Ying and the reason that follows from it. It runs no new computation. Every number below is quoted from an existing record, named with its sha256 where one is known.

This is r2 of `REASON_L1_1_one_BCC.md` (`/workspace/L1_1_one_bcc_reason_20261112/`, sha256 `82e58c43d17f7cafe1f5d6e43d8367e22aaa6fac88590bca5d1052ed018c117e`). That note quoted its paper lines from a stale box copy of the paper, not from main, and it called main's four-taste sentence unapplied, which is false. For every paper citation, this r2 note supersedes it. Every paper citation here refers to main: repository `edmondying-hash/spacetime-ledger`, commit `8251f96564a3394ce093a5260a0d307bb4e2f637`, `space-ledger-v2.md` sha256 `e86ef5c3f1ad9331d61c6033e300dfd37232f4836ce39a55b206bf3985335dc8`. The physics content of the earlier note is kept wherever it is still correct.

1. The choice

In the chat of 12 November 2026, Edmond Ying wrote, in order:

- t2542u: "ok, let's move the less is more approach to one bcc orbit"
- t2543u: "ok, changing this will fix L1.1"
- t2544u: "ok, go ahead and write the reason"
- t2546u, on the paper edits: "yes, please go ahead and make all changes"

The choice is that the physical lattice of the ledger is **one** body-centred cubic (BCC) orbit, the one through the origin: the sites of the integer grid whose three coordinates are all even or all odd. The other three orbits of the integer grid are not part of the physical state space. This is a choice about what the lattice is. It is not a repair of the rule, it changes no symbol, and it is not a new run.

2. Sources

All files were read on the box, read-only, and hashed with `sha256sum` when this note was written.

Document	Path	sha256
Live plan, `layered_plan_v3.3.md` (v3.3 r2)	`/workspace/layered_plan_v3_3_r2_20261111/`	`89b3972a0af418bfcb260b978842e79dc73945c61ec2bd85fe3a5e7d1785aa39`
L0 canonical-rule verification, `VERIFY.md`	`/workspace/L0_canonical_20261109/`	`02945be65e529feb5d7cd10b70e566e3d49e3ab474dc0dec86098348be0ce179`
Official V5 v2 census, `results.json`	`/workspace/L0_V4V5_runs_20261111/v5v2_official/`	`ca6793ea0f2a245c556aabcd65373313fc751902cfb799ea962256e22d212708`
V5 v2 census code, `v5_census.py` (as pinned in that `results.json`)	`/workspace/L0_V5v2_code_r2_20261111/`	`7c6f7ed5aaa6c8fbfe5c823170e1ef3e2c60b3771b9c1406d8bcda394f8f5850`

Document	Path	sha256
Paper on main, space-ledger-v2.md (commit 8251f96564a3394ce093a5260a0d307bb4e2f637)	edmondying-hash/spacetime-ledger, main	e86ef5c3f1ad9331d61c6033e300dfd37232f4836ce39a55b206bf3985335dc8
Stale box copy of the paper, not main (cited by the earlier note)	/workspace/whitepaper/space-ledger-v2.md	53f5050660f3fe11e86e033c5618d82fe639d1b18ed70f0c1a703ec498497944
v3.2 §6.2 text, space-ledger-v2.md.v3.2 (its line 518 is byte-identical to main's line 518)	/workspace/paper_62_draft_20261109/	1c88f4a3baa756c2074c86a7ac369ef7c28d62682693088f2485eee076db4678
Paper edit set, PAPER_EDITS_one_BCC.md	/workspace/paper_one_bcc_edits_20261112/	a6fa09accdae4e89f070a4bb863a4496278e74f096e4bda0f546576ce6c4a167
Its patched paper, main_patched.md	same folder	8cbb94ec481b3332f074ea028505478dd2d576a8b5d60915f072b653ab6aad70
Its diff, main_vs_patched.diff	same folder	ab2f323d2e71c05c4efe9a06c6d69eb268e036e130eb8cb3b46072d9274430dd
Its SHA256SUMS	same folder	05639f944b2f4a43fd79dc2b8c887f3297f51744a2f594cfb27fcac5ef6578f3
r = 0.05 chain operator, run_chiral_wall_count.py (U_slab)	/workspace/chiral_wall_count_20261109/	aa5d0cf86a660594a04aa4157b2dff90fe7a0cac9341406d2bbdae30e1eb2c2c
r = 0.05 results, results_low_r_followup.json	/workspace/chiral_rscan_20261109/	0f16b0811f88bb105ed299015b5d1d6864ecba2e0afb7697c56f00e0408533f4
L1.1 repair specification r8, REPAIR_SPEC_L1_1_r8.md	/workspace/L1_1_repair_spec_r8_20261111/	ad02c20147677e527fc81f5f3b6e27a0dd2f1b300cf8dcde4152cc8fdab63afc
r8 sign-off record, L1_1_repair_spec_r8_20261111_SIGNOFF.txt (bigbang /home/eying/spacetime_files/; box copy in /workspace/ hashes the same)	see left	00c9a8707c4d9880d06a92bd5bb432a8e651dcd4399aa0aa9ad0759813cdf154
Generating stack, qca/dirac4.py (scan script with dist_gamma)	/workspace/qca/	9f01ea8f10947eb6a30506e1499de22e373fc7e1a25fa2689e38c873e2b5aedd
Generating stack, qca/dirac4s.py	/workspace/qca/	4db41fe71ce2844f58f1c61c3c84a768b37b82de1bd546a468d3980c3f0627b4
Generating stack, qca/sym.py	/workspace/qca/	7b1bbafd4b01464c8f4d4643582c60994f34ed3e119c4167625f2e5905131764
Generating stack, qca/dw.py	/workspace/qca/	769833f0393dc1599d5944f6a2837c6ff5ac207ce6ca169eaa2ba9b2ca5fd651
Chain global search, qca/dw_gscan.py	/workspace/qca/	5bf6af5d089fd17dbe922f335855e9bf961dc731be4025b591e3e0ff16bf938e

Document	Path	sha256
Frozen symbol module `dirac4.py` (the one `VERIFY.md` checks; for example `/workspace/chiral_rscan_20261109/src_readonly/dirac4.py`)	see left	`f71c782ed7089680e4c8e2441d12a926de9b8c18b57a9a28253b018ebbab8aff`
OP4 production source, `op4_local_gpu.py`	`/workspace/gpu/op4_bcc_local/loaded_packet_L320/`	`e2164af661b9993b3ec588b06c4969a71333ba35d72e59ec720fcaa363c5cfb3`
Real-space checks, `qca/dirac4_checks.py`	`/workspace/qca/`	`4e316d9895f035966bb0aa269216dcc38c3986c50c5118907ab609af8e519c05`
L1.2 pre-registration, `PREREGISTRATION.md` (staged copy found on the box)	`/tmp/l12_prereg_stage/`	`733752444dfdbaab68654e583b15fdc97619883224fc30357f8ecba998747239`

Two file names need care. The `qca/dirac4.py` in the generating stack (`9f01ea8f…`) is a scan script that defines `dist_gamma` and `global_min`; it has no docstring. The module with the docstring that says hop vectors are `(±1,±1,±1)/2` is the frozen symbol module `dirac4.py` (`f71c782e…`), line 6. Section 6 refers to that one.

3. What L1.1 asks

The plan's L1.1 heading is "L1.1 Doubler census — OP3 (both rules)". Its pass clause reads, exactly:

> **Pass.** For each rule, either no zero except Γ in the closure, or a documented statement that the extra zeros are excluded from the physical state space, and why. **The same decision rule applies to both findings below:** repair it, or give a reason it is unphysical that survives a finite lattice with even L.

The two findings it refers to are the twelve exact zeros of the §6.2 symbol at the face centres `(±π,±π,0)` and permutations (the plan calls them face centres; main's §6.2 calls the same twelve labels "edges, with two components at ±π", line 518), with U = I to 7e−16, and OP4's three extra nodes H = (π,0,0), P = (π/2,π/2,π/2) and P′ = −P. The plan's "Decision needed" paragraph says "one decision, applied to both rules". The L1 exit gate reads "L1.1 decision made and census re-run for the chosen rule". The plan names `dirac4` as the canonical rule and OP4 as a labelled diagnostic branch (plan, L0 status line).

4. Why the twelve face-centre labels are Γ

The facts in this section are from `VERIFY.md` unless another source is named.

1. **The hops are integer.** The formula the code implements, $u = \prod\cos k_i - \varepsilon\prod\sin k_i$ ($\varepsilon = \pm 1$ is a sign), is the eight-corner hop sum for integer hops $h = (\pm1,\pm1,\pm1)$ in the same units as k. OP4 agrees: its eight hop keys are the integer corners, and its real-space bond offsets are ±1.
2. **Four orbits.** Hops $(\pm1,\pm1,\pm1)$ on the integer grid $\mathbb{Z}^3$ split the walk into four decoupled BCC orbits: (0,0,0)↔(1,1,1), (1,0,0)↔(0,1,1), and the two cyclic partners of the second. Each hop flips the parity of all three coordinates, so an orbit is never left. At r = 0 the measured leak between orbits is about 1e−32 of the norm.
3. **Zone volume.** For integer hops the primitive BCC zone has volume $2\pi^3$. The cube $[-\pi,\pi]^3$ has volume $8\pi^3$, which is four times the primitive zone.

4. **Reciprocal vectors.** π(1,1,0) and its permutations are reciprocal vectors of the hop lattice, which is the FCC lattice generated by π(0,1,1) and cyclic. The symbol is invariant under them: the largest difference over 200 random k is 3.51e−16 on the three generators and 8.01e−16 over all twelve sign patterns.
5. **U at the twelve labels.** At the twelve labels of type (±π,±π,0), U equals U(Γ) to 6.83e−16.

The census code uses the same set. Its Γ images are the points πg with g an integer triple of even sum (`v5_census.py`, `GAMMA_IMAGES`; `qca/dirac4.py`, `dist_gamma`, which skips g when `sum(g)%2` is odd). Those even-sum points are exactly the reciprocal lattice of the orbit through the origin.

On one BCC orbit, therefore, the twelve face-centre labels name the point Γ itself, not twelve additional points. The plan's own L0 status line already records that "those twelve points are Γ images under the even-parity lattice and lie in the V5 exclusion set".

5. Why the first pass clause is met for `dirac4`

With one BCC orbit as the physical lattice, the zeros of `dirac4` in the closure are as follows.

- **The 27 high-symmetry labels.** Mod 2π they are 8 distinct momenta. On one orbit they fall into two classes, not four: Γ, and the H class (the faces and the corner). The H class has gap π/2 − φ, which is 0.17079632679489665 at φ = 1.4. Γ is the only zero among them.
- **The interior.** The official V5 v2 census is unrepaired `dirac4` at m = 0, run on a 129^3 boundary-inclusive grid and from 300 random starts, with local minima refined by Nelder–Mead and a 0.3 exclusion around every even-parity Γ image. It reproduced Table `dirac4`. Outside the exclusion it found only a nonzero gap:

φ	smallest non-excluded ω (search a)	distance of argmin to nearest Γ image
0.5	0.217127955311918	2.2214
0.8	0.26167630634482675	2.2214
1.0	0.23202826004369173	1.8901
1.2	0.1668769618006517	1.8616
1.4	0.08212694658054251	1.9360
1.5	0.034884422707531004	2.0266

At φ = π/2 the census value is 1.4e−17, in line with the paper's table entry "0 (doublers return)". That endpoint is not a pass, and it is not the canonical setting. The paper's default is φ = 1.4.

- **Inside the exclusion balls.** Every ball is centred on a Γ image, so on one orbit every ball is a neighbourhood of Γ itself. Near Γ the band is the Dirac cone, with slope ±0.98910736 (`VERIFY.md`, Richardson).

So, for 0 < φ < π/2 at the tabulated values, `dirac4` on one BCC orbit has no zero except Γ in the closure. This is the **first** pass clause of L1.1. It does not rely on the second clause, the "documented exclusion".

The even-L condition. The plan asks that a reason survive "a finite lattice with even L", because k = π lies on the momentum grid of an even lattice. On one orbit this holds for the following reason. Take a periodic cubic grid of even side N, as in the real-space runs (N = 48 or 128 in `dirac4_checks.py`, N = 40 in `dw_real.py`). Wrapping by an even N preserves coordinate parity, so the four orbits stay separate on the torus, and the orbit through the origin is itself a

periodic BCC torus with $N^3/4$ sites. For every site x of that orbit, $\pi(1,1,0)\cdot x = \pi(x_1 + x_2)$ is a multiple of 2π, so the plane wave at k + π(1,1,0) is the same function on the orbit as the plane wave at k. The label (π,π,0) is on the momentum grid when N is even, as the plan says, but on the orbit it is the same state as Γ, not a second state. The finite even lattice therefore leaves the identification intact.

6. Wording in the paper

These are proposals only. This note does not edit the paper. The exact line-by-line changes are in the paper edit set, `/workspace/paper_one_bcc_edits_20261112/` (`PAPER_EDITS_one_BCC.md`, sha256 `a6fa09accdae4e89f070a4bb863a4496278e74f096e4bda0f546576ce6c4a167`; `SHA256SUMS` sha256 `05639f944b2f4a43fd79dc2b8c887f3297f51744a2f594cfb27fcac5ef6578f3`). It changes 14 lines of main: 57, 518, 519, 520, 521, 523, 543, 552, 572, 581, 588, 590, 616 and 623.

Main already carries the four-taste wording. Main's §6.2 "Doublers" bullet, line 518, reads in part: "This is the four-taste cubic-grid rule: the grid is $\mathbb{Z}^3$, the diagonal hops $h = (\pm1, \pm1, \pm1)$ split it into four decoupled BCC orbits, at $r = 0$ there are four cones — Γ plus the three two-π momenta, one per taste — and the cube $[-\pi, \pi]^3$ is four times the primitive $2\pi^3$ zone." This line is byte-identical to line 518 of `space-ledger-v2.md.v3.2` (`1c88f4a3…`), so that text is applied on main. The earlier note called it unapplied. That was wrong. The same reading appears on main at lines 543, 552, 572, 588 and 616. The edit set replaces all of them with the one-orbit reading.

The units sentence is on main **line 523**, not line 524 as the earlier note said: "*Links.* The 8 corner links are $h \in \{(\pm1, \pm1, \pm1)\}$ in units of $a/2$, with $T_h\psi(x) = \psi(x - h)$." The same "units of $a/2$" phrase opens line 518, which also writes the hop sum as $8\prod_i \cos(k_i a/2)$. The sentence is right only if k is also read in radians per a/2. If k is read per a, the cubic cell, the hops become (±1,±1,±1)/2. That gives a 4π period and a zone of volume $16\pi^3$ that the cube only half covers, and it makes (π,π,0) a zone-boundary point distinct from Γ. `VERIFY.md` refutes that reading, because the symbol is invariant under π(1,1,0) (residual 2.05e−15 for integer hops against 2.0 for half hops). The identification in Section 4 depends on the hops being integer in the units of k. With integer hops, the hop sum is $8\prod_i \cos k_i$, matching line 525 ($c_i = \cos k_i$). The edit set changes line 523 to the sentence proposed earlier: "The 8 corner links are h∈{(±1,±1,±1)} in integer units of a/2, and k is measured in radians per a/2, so every symbol below is 2π-periodic in each k_i. The cores occupy one BCC orbit of this integer grid, the sites whose three coordinates are all even or all odd. Then π(1,1,0) and its permutations are reciprocal lattice vectors, the momenta (±π,±π,0) are the point Γ, and the cube $[-\pi,\pi]^3$ covers the primitive zone, of volume $2\pi^3$, four times." It keeps T_h as well, and it rewrites line 518 to match.

§1.2 is main **line 57**, not line 58: "the body-centred cubic (BCC) grid with 8 links per node". This already agrees with the choice. The edit set adds "(one BCC orbit of the integer cubic grid: the sites whose three coordinates are all even or all odd, joined by the corner links $(\pm1, \pm1, \pm1)$)".

§6.2 low-energy limit is main **line 543**, not line 544. Main no longer says "the only zero-frequency point is k=0". It says: "Besides Γ, a census at $\varphi = 1.4$ finds three distinct momenta (twelve sign labels), the edges with two components at $\pm\pi$ (see Doublers, above)." The edit set replaces this with: at the tabulated values $0.5 \le \varphi \le 1.5$, a census finds no zero-frequency point other than k = 0 (up to reciprocal lattice vectors), because the twelve edge labels are Γ itself on the one-orbit lattice, and at φ = π/2 the doublers return.

The frozen `dirac4.py` docstring (`f71c782e…`, line 6) gives "hop vectors h = (+-1,+-1,+-1)/2". That is wrong for the code, which uses integer hops. The file is frozen and is not edited.

Plan frozen conventions (plan line 100) read "BCC links $h = (\pm1,\pm1,\pm1)/2$, $\ell = a\sqrt{3}/2$". These have the same dependence on the unit of k. This is flagged here and not edited.

7. Consequences

- **Sum-form terms are off the table.** The standard Wilson term $r\,\Sigma_i(1-\cos k_i)$ and the face-centre patch $g(k) = (1/3)\,\Sigma_i(1-\cos k_i)$ (frozen `dirac4.py`, `_u_n` with `patch` at line 37, and `S_step` with `wilson` at line 66) are not periodic under π(1,1,0). Their axis hops of length 1 leave a BCC orbit. `VERIFY.md` measures this: at r = 0.05 the sum form gives t = 0.2 at the edge copies against 0 at Γ, and about 0.0022 of the norm leaks to each other orbit. On one orbit these terms are not lattice terms.
- **Main's r = 0.05 repair uses the sum form.** Main lines 519 and 588 adopt a chain gap of 2.663×10^{-2} E_P, with a Wilson term of mass r = 0.05, on the interior 41^3 grid at N_s = 32. That number is `results/0.05/by_Ns/32/interior41/omega_min` = 0.026625477907575197 in `results_low_r_followup.json` (`0f16b081…`). It was computed through `U_slab` (`run_chiral_wall_count.py`, `aa5d0cf8…`, lines 184–195), which passes `wilson_mass` to frozen `dirac4.py` `S_step` line 66, `t = arcsin(m) + wilson·Σ(1 − cos kᵢ)`. `VERIFY.md` line 59 also names it "standard Wilson … the adopted 2.663×10^{-2}". A check in the edit set's folder (`one_bcc_checks.py`, `f6418b41…`) gives a shift difference of 0.2998 under π(1,1,0) at r = 0.05, against 1.1×10^{-15} at r = 0. The edit set therefore withdraws the repair and every number derived from it from lines 519 and 588, without replacing them. It keeps the r = 0 numbers, which do not depend on it.
- **The product form stays.** $r(1-\prod\cos k_i)$ uses only on-site and the eight corner hops, stays inside one BCC orbit, and keeps π(1,1,0) periodicity (largest Δt 2.78e−17).
- **Real-space packets.** The Gaussian packets in `dirac4_checks.py` sit on the full cubic grid and put 0.2500 of the norm on each of the four orbits (σ = 4 and σ = 8). Under the choice, such a run reads as four decoupled copies of the one-orbit rule at r = 0. A point source at the origin, as in the unitarity and light-cone tests and in `dw_real.py`, occupies one orbit already.
- **Table `dw` and the L1.2 pre-registration are unaffected.** The chain stack (`dw.py` → `dirac4s.py` → `dirac4.py` → `sym.py`) builds every layer from $\cos k_i$ and $\sin k_i$ of the corner hops, plus on-site and inter-copy mixing, so it is periodic under the BCC reciprocal vectors. Its exclusion (`dist_gamma`, used by `dw_gscan.py` and named in the L1.2 pre-registration as "even-parity Γ images, which include the face centres") already removes exactly the Γ images of one orbit. So the r = 0 chain gaps on main line 588 (0.058, 0.046, 0.034 and 0.022 E_P at N_s = 16) and the table values on main lines 558–563 stay valid as doubler gaps of the one-orbit rule. Main's statement that they are "not a global minimum while r=0" falls away, and the edit set removes it. The doubler wall mode that survives for $b \geq 0.3$ near $(\pi/2,\pi/2,\pm\pi/2)$ is a genuine BCC point. It differs from Γ by a vector that is not a reciprocal vector, so the choice does not remove it, and the paper's requirement $b \leq 0.25$ stands.

8. Status of r7, r8 and the additive route

The choice is recorded as a lattice choice. It is not a repair, and nothing was run to make this note.

- `REPAIR_SPEC_L1_1_r8.md` (`ad02c20147677e527fc81f5f3b6e27a0dd2f1b300cf8dcde4152cc8fdab63afc`) stays hashed and signed off, per its sign-off record

(`00c9a8707c4d9880d06a92bd5bb432a8e651dcd4399aa0aa9ad0759813cdf154`). `REPAIR_SPEC_L1_1_r7.md` (`aeca56e4a60bf57a929bb48b1651a23863651e97b41ebeb1b87a06663380ebbb`) is unchanged.

- Under the choice, r8 is no longer the required L1.1 path for `dirac4`, because the first pass clause is met without a repair. Nothing in r8 is withdrawn. Its findings stand as recorded, including the closure of the additive route (routes 1 and 2) by exact unitarity.
- This note does not decide the §3.4 floor of r8, "4/4", or what P is for.

9. OP4 is not covered

OP4 production stores one BCC orbit: packed all-even A and all-odd B sites, with no weight on the other three orbits. Its symbol has four isolated Weyl nodes per chirality block, at Γ, H = (π,0,0), P = (π/2,π/2,π/2) and P′ = −P, with chiralities (+,−,+,−) (plan, L1.1). None of H, P or P′ differs from Γ by a reciprocal vector, so all three are genuine points of one BCC zone. The one-orbit choice does not remove them.

OP4 is the labelled diagnostic branch, so this does not block the canonical gate. But the plan's L1.1 is written for "both rules", and that requirement still needs a separate reason or a repair for OP4. This note does not claim that OP4 passes L1.1.

10. What remains for Edmond

1. Whether to apply the paper edit set (`/workspace/paper_one_bcc_edits_20261112/`, 14 lines of main) to the paper, and the matching L1.1 line (and the frozen-conventions line) to the plan. The edit set lists its own open items, including which length is ℓ_P (a or a√3/2) and the figure images, which are not regenerated.
2. How the plan's "both rules" requirement is handled for OP4: a reason, a repair, or a change to the requirement.
3. Whether the existing V5 v2 census is accepted as the census for the chosen rule under the L1 exit gate wording ("census re-run for the chosen rule"), or a new census is wanted. This note does not run one.